SAR Image Segmentation Based on Complicated Region-Sensitive Adaptive Superpixel Generation and Hybrid Edge Correction

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***Abstract*— Superpixel segmentation algorithms are predominantly based on simple linear iterative clustering (SLIC), and treat homogeneous and complex regions equally. This can lead to suboptimal segmentation results, especially in complex images with multiple objects. We address this problem by proposing an SAR image segmentation algorithm based on complicated region-sensitive adaptive superpixel generation and hybrid edge correction (RSASGEC). First, a dynamic initialization algorithm for superpixel seeds based on region complexity is designed. Specifically, a new superpixel representation structure for superpixel seeds is constructed by combining superpixel complexity and the number of contained pixels. The algorithm gives priority to regions with high complexity, dynamically selecting the region with the highest complexity for further partitioning. This results in a dense distribution of superpixel seeds in complex regions, and sparse distributions in homogeneous regions with low complexity. Second, an iterative superpixel segmentation process based on an adaptive energy function is proposed. The Lagrange multiplier mathematical strategy is employed to optimize the adaptive energy function within an adjustable search window, resulting in more compact superpixel segmentation. Finally, a label correction method, based on edge mixture model constraints, is proposed for postprocessing. By integrating edge information from the Gaussian edge detector and the Canny algorithm as constraints, this method leverages majority voting and region growth methods to mitigate edge noise and outliers, refining the superpixel labels. The RSASGEC algorithm is verified in experiments, using one simulated image and six real SAR images. The results indicate that RSASGEC outperforms six representative algorithms, achieving more satisfactory segmentation performance.**

***Index Terms*— Adaptive superpixel generation, dynamic initialization, edge correction, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active Earth observation imaging technology known for its all-day, all-weather, and high spatial resolution characteristics [1]. However, the speckle noise caused by the image backscattering mechanism [2] poses a significant challenge in SAR image interpretation [3]. SAR image segmentation is fundamental for interpreting remote sensing images, playing a crucial role in subsequent tasks like ground object detection [4], [5] and scene understanding [6], [7].

SAR image segmentation methods can be typically divided into deep learning-based methods [8], [9] versus more conventional methods using hand-crafted features [10], [11]. Deep learning methods use hierarchical structures to extract image features automatically, reducing manual dependencies [12]. For example, Yue et al. [13] proposed an attention-based deep model that enhances segmentation accuracy by incorporating multiscale attention and a well-designed loss function. Zhang et al. [14] introduced a domain adaptation technique based on generative adversarial networks for segmentation. Ma et al. [15] presented a network that integrates superpixel generation and segmentation within a deep end-to-end framework. Although these methods achieve impressive performance, they rely heavily on labeled data for training. However, labeling SAR image is a labor-intensive process that demands considerable professional expertise. This bottleneck limits the widespread adoption of deep learning methods, making conventional methods still valuable for SAR segmentation.

In conventional SAR image segmentation methods, superpixel methods [16], [17] have attracted increasing attention. Superpixels act as an intermediate representation in the segmentation process, offering a means to capture spatial context information within an image [18]. They group pixels with similar properties into perceptual patches [19], effectively reducing the computational load by working with larger units. Within each superpixel, there is a high consistency in color, texture, or other features [20], leading to more coherent segments. Compared to pixel-wise methods [21], [22], superpixel-wise methods demonstrate superior computational

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efficiency while preserving the image's structural integrity and enhancing boundary adherence [23]. The precision of generated superpixels significantly impact the accuracy of SAR image segmentation. Finer superpixels tend to yield more satisfactory segmentation results [15].

Superpixel generation methods can be divided into graph-based and gradient-based methods [24]. In graph-based methods, image pixels are treated as nodes in a graph, where the similarity between pixels determines the edge weights. Graph-based methods include Normalized Cutting (N-Cuts) [25], Entropy Rate Superpixel segmentation (ERS) [26], and so on. Xiang et al. [27] devised an enhanced N-Cuts strategy by combining K-means clustering, N-Cuts, and edge constraints, resulting in superpixels with precise boundary adherence. However, optimizing graph parameters entails high computational complexity. In contrast, gradient-based methods [16] typically initialize superpixel centers coarsely, and then refine them iteratively using gradient ascent. One particularly effective and representative gradient-based method is the simple linear iterative clustering (SLIC) algorithm [28].

SLIC, initially developed for optical images, involves three main steps: initialization, local K-means clustering, and postprocessing. During initialization, superpixel centers are placed by sampling pixels on a regular grid across the image. In the subsequent local clustering step, weighted Euclidean distance is utilized to assess pixel similarity in intensity and spatial dimensions. The K-means algorithm is then applied within a fixed search window to update the centers of superpixels [28]. Despite its ease of implementation, memory efficiency, and conceptual simplicity, SLIC has demonstrated superior performance in optical image segmentation [29]. However, the presence of inherent multiplicative speckle noise in SAR images introduces random variations in pixel intensity, making the additive model unsuitable for SAR intensity modeling.

In recent years, researchers have continuously improved the similarity measure in the SLIC algorithm to better capture SAR statistical features. Xiang et al. [20] pioneered the SLIC algorithm for SAR image superpixel generation, introducing a distance measure that considers both intensity and spatial information. Qin et al. [30] developed a grid-centered initialization strategy (SLIC-GC) and designed the Wishart hypothesis test distance tailored for PolSAR images. Akyilmaz and Leloglu [31] combined Mahalanobis distance (MD) with a similarity ratio to create a linear distance metric, enabling superpixels to adopt stretchable shapes and better adapt to complex structures. Zou et al. [32] employed the generalized gamma distribution (GΓD) to model SAR images, and integrated SAR image likelihood information into the similarity measurement. Considering the generalized likelihood ratio (GLR)'s potential in handling multiplicative noise, Guan et al. [33] designed a GLR-based similarity metric for superpixel generation. Jing et al. [34] defined an edge-aware dissimilarity measure, which incorporates edge information. Jing et al. [23] proposed a new dissimilarity measure with an edge penalty and a trade-off factor to balance spatial and GLR dissimilarity, thereby maintaining boundary adherence. However, the SLIC algorithm does not enforce pixel connectivity, which can result in isolated pixels after clustering.

Edges information can significantly reflect changes in the image's statistical properties, and is increasingly being integrated into postprocessing to ensure superpixel connectivity [35]. Hu et al. [36] introduced edge information and proposed an edge-dominated SLIC clustering method for SAR image segmentation. Jing et al. [34] adopted a scale-based edge detector to develop an edge-sensitive superpixel generation method, improving superpixel connectivity and boundary adherence. Shang et al. [37] proposed a superpixel boundary-based edge description algorithm (SpBED), which combines weak edges from SLIC with strong edges detected by the edge detector to enhance SAR segmentation. Zhang et al. [19] presented an edge intensity map extracted by the edge extractor to measure pixel dissimilarity, making the segmentation results more compact. Additionally, Pappas et al. [22] proposed the generalized gamma mixture model superpixel (MISP-GΓD) algorithm, which uses the generalized gamma distribution to enhance superpixel boundary adherence. Shang et al. [38] integrated the Fisher vector into superpixel segmentation, correcting noise in clustering results with edge information to effectively segment SAR regions with low gray contrast.

The improvement of SLIC in the aforementioned algorithms primarily focuses on refining the similarity distance in local clustering and postprocessing stages, with comparatively fewer improvements in the initialization step. Qin et al. [30] developed SLIC-GC, which employs a grid-centered (instead of seeds-centered) clustering strategy to prevent initial cluster center deviations. Based on SLIC-GC, Yin et al. [39] further improved the initialization method by incorporating image gradient information to guide spatially constrained watershed (SCoW) labeling, resulting in a more robust initialization of superpixel centers. However, the existing initialization methods often obtain superpixel center seeds through regular grid sampling, typically at intervals of S, with subsequent clustering searches typically confined to the $2S \times 2S$ region. These methods may overlook the complexity of different regions, treating homogeneous and complex areas equally. Consequently, when processing complex regions involving multiple targets, challenges arise in capturing texture structure and effectively handling boundary blurring, leading to suboptimal segmentation. Thus, there remains much room for improvement in superpixel generation.

To address this issue, this article proposes complicated region-sensitive adaptive superpixel generation and hybrid edge correction (RSASGEC). First, a novel complicated region-sensitive dynamic initialization method is presented. This method involves constructing a new representation for superpixels with complexity. Unlike conventional sampling on regular grids with fixed intervals of S, this method dynamically allocates initial superpixel seeds based on the region's complexity. Specifically, it allocates more superpixel seeds in complex regions and fewer in homogeneous regions. Second, an adaptive energy function is designed, which combines similarity measurement and adaptive factors. Within the adjustable search window, the Lagrange multiplier mathematical strategy is employed to optimize the adaptive energy function, achieving a more compact superpixel

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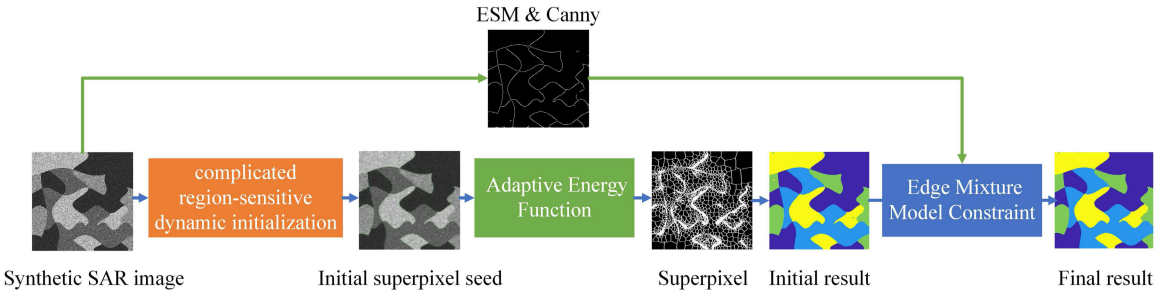


Fig. 1. Overall framework of the proposed RSASGEC algorithm.

segmentation. Finally, a label correction method based on an edge mixture model is proposed. Specifically, it integrates edge information extracted by the Gaussian edge detector and the Canny algorithm as constraints. Furthermore, majority voting and region growth methods with edge constraints are incorporated to effectively mitigate the problem of edge noise and outliers.

The contributions of this article are as follows.

1) A dynamic initialization method for superpixel seeds based on region complexity is proposed. This method dynamically partitions superpixels according to their complexity, allowing the initial superpixel centers to better reflect the structure of the image.
2) An adaptive energy function with adjustable factors is designed and optimized using the Lagrange multiplier strategy. It adjusts automatically within an adaptive search window based on the image's local characteristics and structure, resulting in more compact superpixels.
3) An edge mixture model is constructed for postprocessing, combining Gaussian edge detector and the Canny algorithm to generate edge information. With the constraint of edge information, labels are corrected based on majority voting and region growth methods, helping to eliminate edge noise and outliers.

II. Proposed Method

Traditional SLIC-based algorithms initially divide the entire image uniformly, and obtain initial superpixels by sampling on a regular grid. When dealing with complex regions in SAR images, there are challenges in capturing texture structures and handling of boundary blurring. Overly large superpixels contain multiple targets that may overlap, touch, and have similar texture features, leading to difficulty distinguishing each target. Conversely, overly small superpixels may fail to fully contain the target, resulting in incomplete target segmentation. To tackle these challenges more effectively, the RSASGEC algorithm is proposed to be sensitive to complex regions in SAR images.

The overall framework of RSASGEC is shown in Fig. 1. In Fig. 1, Gaussian smoothing is first performed on the input image to reduce noise. Then, a novel complicated region-sensitive dynamic initialization method is designed to initialize the superpixel seeds. This method involves constructing a new structure for superpixels, which intuitively represents superpixel complexity, coordinates, and the number of pixels. The initial superpixel seeds are generated by dynamically segmenting the region with the highest complexity. These superpixel seeds are distributed more densely in regions of high complexity, and more sparsely in homogeneous regions. Next, the initial superpixel seeds undergo local clustering to generate superpixels. In this process, an adaptive energy function with adjustable factors is designed to measure the similarity between pixels and superpixels, within an adjustable search window. A Lagrange multiplier strategy is employed to optimize this function. Finally, an edge mixture model is utilized for label correction for the generated superpixels to obtain the final result. This final step helps to eliminate edge noise and outliers.

A. Complicated Region-Sensitive Dynamic Initialization

SLIC and its variants typically select seeds according to a regular grid for superpixel initialization, and then perform local clustering within a region with a fixed search window size. However, this often leads to superpixels with uniform sizes, which may not adequately represent the features of the image. Consequently, superpixels may encompass multiple distinct targets in complex regions. To overcome this limitation, the complicated region-sensitive dynamic initialization algorithm is introduced. Instead of adhering to a fixed grid for initialization, the RSASGEC algorithm dynamically selects initial superpixel seeds based on the region's complexity. It prioritizes complex regions during initialization, ensuring that superpixels align with the structure and content of the image more effectively.

In the initialization stage, the superpixel region is a square and has a regular shape. The superpixel can be described by the boundary coordinates, and superpixel center coordinates can be calculated from the bounding box coordinates. To provide a more intuitive representation of superpixel complexity, a novel representation of the superpixel, denoted as α, is constructed. This representation incorporates complexity, index, the number of contained pixels, and the coordinates of the superpixel. This facilitates subsequent superpixel center initialization based on complexity. Specifically, assuming that the image contains N pixels and is eventually segmented into L superpixels, the expression of α is as follows:

$$\alpha = \left[E, l, n, x_l, y_l, x_u, y_u\right] \tag{1}$$

where E represents the average complexity of superpixels, l represents the superpixel index, and n represents the

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number of pixels contained in each superpixel. The coordinates $[x_l, y_l, x_u, y_u]$ define the bounding box coordinates of each superpixel. x_l and y_l represent the horizontal and vertical coordinates of the lower left corner of the superpixels, respectively, while x_u and y_u represent the horizontal and vertical coordinates of the upper right corner of the superpixels.

The calculation for the average complexity of the ith superpixel region is as follows:

$$E_i = \frac{\sum_{j=1}^{|S_i|} \left(I_j - \bar{I}_i\right)^2}{|S_i|} \tag{2}$$

where $|S_i|$ represents the number of pixels in the ith superpixel. I_j is the gray value of the jth pixel. $\bar{I}_i$ is the mean value of the ith superpixel.

Next, the complexity of the regions is sorted in descending order. The region with the highest complexity is dynamically selected and further subdivided into four blocks. This process is repeated until the number of superpixels reaches L. By continuously dividing the region with the highest complexity, superpixel regions based on complexity are achieved. The central coordinate of the superpixel region is the position of the superpixel center point

$$\begin{cases} x_c = \dfrac{x_l + x_u}{2} \\ y_c = \dfrac{y_l + y_u}{2}. \end{cases} \tag{3}$$

As the iteration progresses, the shape of the superpixel changes according to the image content, and gradually becomes irregular. The bounding box coordinates $[x_l, y_l, x_u, y_u]$ in (1) can no longer precisely describe the superpixel range. To accurately represent the superpixel center, instead of using $[x_l, y_l, x_u, y_u]$, we update the superpixel center by taking the average of the coordinates of all pixels within the superpixel, i.e., the geometric centroid

$$\begin{cases} x_c = \dfrac{\sum_{i=1}^{n} x_i}{n} \\ y_c = \dfrac{\sum_{i=1}^{n} y_i}{n}. \end{cases} \tag{4}$$

This center coordinate is used for superpixel segmentation and update in the next iteration. The representation of each superpixel center point, denoted as α_c, is obtained as follows:

$$\alpha_c = \left[E, l, n, x_c, y_c\right]. \tag{5}$$

The process of superpixel initialization is shown in Fig. 2.

In Fig. 2(a), the superpixel initialization method of SLIC evenly divides the rows and columns of the input image based on a predefined number of superpixel seeds. This regular initialization method leads to the emergence of multiple targets in the generated superpixels, posing a challenge for effective segmentation. Fig. 2(b) illustrates the complicated region-sensitive dynamic initialization of RSASGEC. It first divides the input image evenly into four equal parts, computes region complexity using (2), and sorts the regions in descending order. The most complex region is then dynamically selected for further partitioning, iteratively continuing until the predefined number of superpixels is reached.

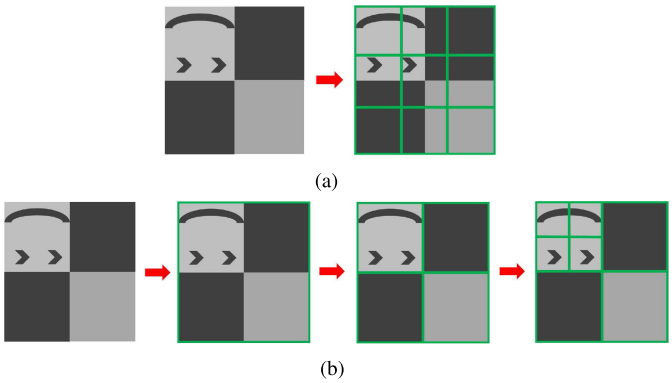


Fig. 2. Process of superpixel initialization. (a) Initialization of SLIC and (b) complicated region-sensitive dynamic initialization of RSASGEC.

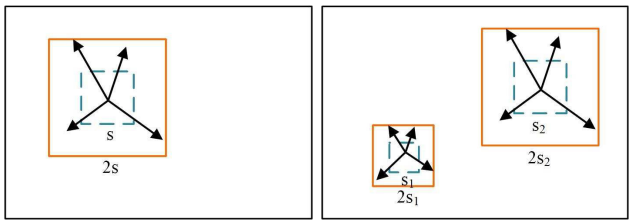


Fig. 3. Local search process. (a) Local search of SLIC and (b) local search of the RSASGEC.

Furthermore, similar pixels are typically searched within a fixed-size window during the local clustering step, commonly set at $2S \times 2S$. However, different regions of the image have different complexity, so it is necessary to design a search window with an adjustable size. Some regions may require a wider search window to capture more complex image features, while others may suffice with a smaller window. This predefined search window size restricts optimal edge adhesion to some extent. Considering that the initial superpixels are densely distributed in complex regions, an adaptive search window based on complexity is designed for the subsequent local search. This adaptive approach ensures better utilization of the varying complexities across the image. The local search process is illustrated in Fig. 3.

Fig. 3 illustrates the process of local search. In Fig. 3(a), it is evident that all search window sizes are fixed at $2S$ for standard SLIC methods. However, in Fig. 3(b), the complexity varies across different locations in the image, leading to adaptive changes in window size. The search window size for any given superpixel is determined by the upper and lower limits of coordinates, as indicated in (1). Adjusting the search window size based on region complexity facilitates better boundary adherence.

B. Iterative Superpixel Segmentation Based on Adaptive Energy Function

Conventional superpixel segmentation methods define a similarity metric or function, and then measure the similarity between pixels and superpixel seed points in a specified region. For each pixel, the greater the similarity, the higher the probability of being identified associated with the superpixel.

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Locally, each pixel's label is labeled as the label of the nearest superpixel seed point. However, with such methods, the boundary pixels often become incorrectly assigned, which increases the total distance of pixels within the superpixel. To address this issue, we propose a superpixel adaptive energy function with adjustable factors. The energy function is optimized using the Lagrange multiplier method. Additionally, the superpixel centers are updated through iterative processes within an adaptive search window. The specific process is as follows.

After superpixel initialization and segmentation, each seed point conducts range detection within a $2S \times 2S$ neighborhood. For each superpixel, the search window size, denoted as $2S \times 2S$, is adaptive and determined by the upper and lower limits of coordinates. When a pixel is closer to the central seed of a superpixel, it is more likely to be assigned to that superpixel. It is important to choose an appropriate similarity measurement to measure the distance between pixels and the center point.

SAR images have inherent multiplicative speckle noise, whose intensity usually follows a gamma distribution. Therefore, the additive noise model (more typical in optical images) does not transfer well to modeling the intensity of SAR images. Recent studies have shown that the GLR can effectively model SAR image statistics. Therefore, GLR is used to measure the intensity similarity of SAR images. Combining intensity and spatial similarity, the similarity distance can be calculated using the following equation:

$$D_w(i,j) = \begin{cases} \dfrac{S_P}{\theta} \cdot \log\left(\sqrt{\dfrac{I_S^i}{I_S^j}} + \sqrt{\dfrac{I_S^j}{I_S^i}}\right), & w = 1 \\ \sqrt{\left(x_i - x_j\right)^2 + \left(y_i - y_j\right)^2}, & w = 2 \end{cases} \tag{6}$$

where w represents the type of distance metric. When $w = 1$, $D_w(i,j)$ represents the intensity distance between pixels i and j. When $w = 2$, $D_w(i,j)$ represents the spatial distance between pixels i and j. S_p is the initial size of superpixels, and θ is the weight control parameter. I_s is the smoothed image, and x_i, y_i represent the horizontal and vertical coordinates of pixel i, respectively. The intensity distance involves the squared difference in pixel intensities, while the spatial distance considers the Euclidean distance between pixel coordinates.

The superpixel adaptive energy function is defined as follows:

$$S = \sum_{w=1}^{K} S_w = \sum_{l=1}^{L}\sum_{i=1}^{N}\sum_{w=1}^{K} (\lambda_w)^{\eta} f_{i,l} D_w(i, C_l) \tag{7}$$

where S_w represents the energy function corresponding to the wth feature. L, N, and K represent the number of superpixels, the number of pixels in the image, and the number of weights, respectively. λ_w is an adjustable weight parameter which adaptively adjusts the importance of two distance metrics. η is defined as the power exponent of the feature weight parameter λ_w, which adjusts the distribution of feature weights in the form of power exponent, to influence the model's ability to distinguish different categories. $f_{i,l}$ represents the membership relationship between the ith pixel and the lth superpixel center, reflecting their degree of association. If the ith pixel does not belong to the lth superpixel, then $f_{i,l} = 0$. $D_w(i, C_l)$ represents the distance between the ith pixel and the lth superpixel center.

A smaller adaptive energy function corresponds to a more compact superpixel segmentation result. Therefore, S needs to be minimized in each iteration

$$\begin{cases} \min, \quad S, \\ \text{s.t.}, \quad \displaystyle\sum_{w=1}^{K} \lambda_w = 1 \\ \quad\quad \displaystyle\sum_{l=1}^{L} f_{i,l} = 1 \quad \forall i. \end{cases} \tag{8}$$

The minimization problem of S can then be reduced to an optimization problem with constraints. The Lagrange multiplier method is a mathematical strategy for solving constrained optimization problems. Its main idea is to transform an optimization problem with constraints into an unconstrained problem by introducing Lagrange multipliers. In this article, we use the Lagrange multiplier method to optimize S. By introducing Lagrange multipliers, the optimization problem is transformed into the saddle point of the Lagrange function.

The Lagrange function is constructed as follows by introducing Lagrange multipliers:

$$L = S + \alpha\left(1 - \sum_{w=1}^{K} \lambda_w\right) + \sum_{i=1}^{N} \beta_i \left(1 - \sum_{l=1}^{L} f_{i,l}\right) \tag{9}$$

where α and β_i are the Lagrange multipliers.

When the gradient of the Lagrange function is zero, S can reach an optimal value. Therefore, the optimization of S is transformed into finding the saddle point of the Lagrange function. The partial derivative of the adjustable weight parameter λ_w can be represented as follows:

$$\frac{\partial L}{\partial \lambda_w} = \eta \lambda_w^{\eta-1} \sum_{l=1}^{L}\sum_{i=1}^{N} f_{i,l} D_w(i, C_l) - \alpha = 0. \tag{10}$$

The expression λ_w can be obtained as follows by solving (10):

$$\lambda_w = \frac{\alpha}{\eta\left(\sum_{l=1}^{L}\sum_{i=1}^{N} f_{i,l} D_w(i, C_l)\right)^{\frac{1}{\eta-1}}} = \frac{\alpha}{\eta (S_w)^{\frac{1}{\eta-1}}}. \tag{11}$$

Substituting λ_w into the constraint of $\sum_{w=1}^{K} \lambda_w = 1$, the expression for α can be obtained as follows:

$$\alpha = \frac{1}{\sum_{w=1}^{K}\left(\frac{1}{\eta (S_w)^{\frac{1}{\eta-1}}}\right)} = \frac{1}{\sum_{k=1}^{K}\left(\frac{1}{\eta (S_w)^{\frac{1}{\eta-1}}}\right)}. \tag{12}$$

Substituting α into (11), the expression for λ_w can be obtained as follows:

$$\lambda_w = \frac{1}{\sum_{k=1}^{K}\left(\frac{1}{\eta (S_w)^{\frac{1}{\eta-1}}}\right)\eta (S_w)^{\frac{1}{\eta-1}}} = \frac{1}{\sum_{k=1}^{K}\left(\frac{S_w}{S_k}\right)^{\frac{1}{\eta-1}}}. \tag{13}$$

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Assuming that the current iteration is T, then the λ_w of the Tth iteration can be expressed as follows:

$$\lambda_w^T = \frac{1}{\sum_{k=1}^{K}\left(\frac{S_w^T}{S_k^T}\right)^{\frac{1}{\eta-1}}}. \tag{14}$$

By bringing (14) into (7), $(T+1)$th energy function S^{T+1} can be updated as follows:

$$S^{T+1} = \sum_{w=1}^{K}\sum_{l=1}^{L}\sum_{i=1}^{N}\left(1/\sum_{k=1}^{K}\left(\frac{S_w^T}{S_k^T}\right)^{\frac{1}{\eta-1}}\right)^{\eta} f_{i,l} D_w(i, C_l). \tag{15}$$

The design of S aims to find suitable superpixel centers. The superpixel center coordinates are derived through the weighted average of the coordinates and weights of each pixel in the superpixel region, as follows:

$$\left(x_{C_l}, y_{C_l}\right) = \frac{\sum_{i=1}^{N} f_{i,l}(x_i, y_i)}{\sum_{i=1}^{N} f_{i,l}}. \tag{16}$$

Based on the updated coordinates of superpixel centers, the image is resegmented to obtain new segmentation results which better adapt to the characteristics of the data.

C. Label Correction Based on Edge Mixture Model Constraints

SLIC does not have strong connections, and there are many small blocks of superpixels in the generated superpixels. Postprocessing can effectively enhance regional connectivity by fusing superpixels. We propose an edge mixture model, based on Gaussian edge detection and the Canny algorithm, to perform class calibration on the generated superpixels. It can effectively prevent the fusion of adjacent superpixels with a significant gray mean difference. Specifically, the Gaussian window weighs each pixel individually to get the final weighted gray. The Gaussian window function is as follows:

$$G(x, y) = \frac{1}{\sqrt{2\pi}\sigma_x\sqrt{2\pi}\sigma_y}\exp\left(-\left(\frac{x^2}{2{\sigma_x}^2} + \frac{y^2}{2{\sigma_y}^2}\right)\right) \tag{17}$$

where σ_x and σ_y represent the standard deviation of the Gaussian window along the x- and y-axes, respectively. $G(x, y)$ is the weight value at the coordinate points (x, y).

To realize multiscale edge detection, we introduce Gabor function templates of different scales. We use three scale templates with sizes of 7×7, 11×11, and 15×15, each with eight orientations, and a total of 24 edge detection templates. After the edge detection template is obtained, the ratio of average (ROA) edge detection algorithm is used to calculate the edge response. The ROA edge detection response is calculated as follows:

$$R^t(x, y) = 1 - \min\left\{\frac{\chi_1}{\chi_2}, \frac{\chi_2}{\chi_1}\right\}, t \in [1, 24] \tag{18}$$

where t represents the index of Gabor edge detection templates. $R^t(x, y)$ represents the ROA edge detection response by the tth template at the coordinate (x, y). χ_1 and χ_2 represent the weighted mean of the symmetrical bimodal Gabor function template. The maximum response of ROA edge detection response is calculated as follows:

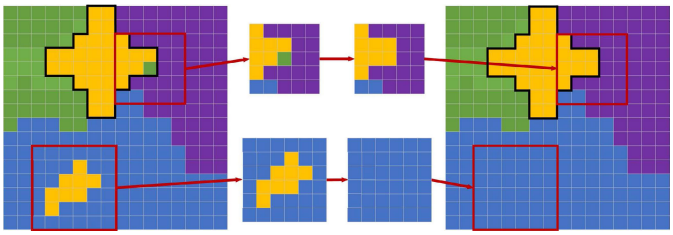


Fig. 4. Process of label correction.

$$M(x, y) = \max\left\{R^t(x, y) \mid t \in [1, 24]\right\} \tag{19}$$

where $M(x, y)$ represents the ROA maximum response by the tth template at the coordinate (x, y). The gradient of ROA edge detection response is calculated as follows:

$$D(x, y) = \arg\max\left\{R^t(x, y) \mid t \in [1, 24]\right\} + \frac{\pi}{2} \tag{20}$$

where $D(x, y)$ represents the gradient of the$t$th template at the coordinate (x, y), respectively.

Subsequently, the Canny algorithm is used for edge detection. The edge information of two edge detectors is fused using the depth-first algorithm, and the edge mixture model edge is obtained. Next, the superpixel is further segmented using the edge constraints. The region growth algorithm is used for the region fusion of superpixels. Region fusion is performed for the superpixel image less than the threshold value, if there is no edge information in the neighborhood. The fusion process is as follows:

$$\bar{I}_i = \begin{cases} \dfrac{\bar{I}_i \times |S_i| + \bar{I}_j \times |S_j|}{|S_i| + |S_j|}, & |S_i| < T \\ \bar{I}_i, & \text{else} \end{cases} \tag{21}$$

where T is the preset threshold for fusing superpixels. The ith pixel group has a neighborhood and no edge limit. If it is smaller than T, superpixel fusion is performed. Otherwise, it continues to traverse the next superpixel.

Traditional K-means clustering is sensitive to noise and tends to converge to local optima, potentially causing clustering errors. There is inherent multiplicative noise in SAR images, leading to segmentation speckles. Meanwhile, smoothing the pixels in the edge area amplifies the difference between high and low gray levels, causing these pixels to be misclassified as a third category. To solve these problems, the improved K-means algorithm is used on the fused superpixels. By combining the edge-constrained majority voting method and the region-growing method, the edge noise and isolated points can be effectively eliminated, thereby correcting class labels. The process of label correction is shown in Fig. 4.

Fig. 4 illustrates the label correction of boundary noise and isolated points. First, the edge-constrained majority voting method is used to eliminate the edge noise. The edge-constrained majority voting method first checks whether a pixel is located within a specific region based on edge detection. If it does, the pixel is assigned to that region. If not, the region it belongs to is determined by the result

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of a majority vote. In the 5 × 5 red box, it can be observed that there is a green pixel in the yellow and purple regions. It is located inside the yellow area, as shown by the black line in the figure. Although the purple pixels dominate the fixed window containing the green pixel, the green pixel is corrected to the yellow pixel due to edge constraints.

Next, the region-growing method is employed to correct the isolated regions. Initially, the region-growing method is used to relabel each pixel. T_s is defined as the threshold value for the pixel number in the superpixel. The yellow pixel block with pixel numbers less than T_s is identified, as shown in the 6 × 6 red box. In this region, the yellow pixel block is misclassified and isolated within the blue pixels. Since this yellow pixel block is not at the boundary and is surrounded only by blue pixel blocks, it is corrected to blue. In summary, the edge-constrained majority voting algorithm can effectively identify edge features in the image. The region-growing method can merge neighboring pixels or regions based on similarity, thereby forming more cohesive regions. By combining them, the improved K-means algorithm enhances the accuracy of image segmentation and effectively handles complex backgrounds and noise interference.

The presence of speckles severely constrains the improvement of SAR image segmentation performance. The RSASGEC algorithm addresses speckle noise impact comprehensively across the entire process. First, in the SAR image preprocessing stage, a Gaussian kernel filter is applied to smooth the image, eliminating the noise. Second, the RSASGEC algorithm emphasizes superpixel generation over individual pixels, effectively mitigating speckle noise across pixels. Moreover, misclassification of superpixels can hinder clustering performance. Therefore, in the postprocessing stage, the RSASGEC algorithm designs a model based on Gaussian edge detection and the Canny algorithm to correct edges, providing further speckle noise mitigation. Overall, the RSASGEC algorithm effectively mitigates the impact of speckle noise on SAR image segmentation performance by combining multiple processes at different stages.

III. Experimental Results and Analysis

To verify the effectiveness of the proposed RSASGEC algorithm, six representative algorithms are selected as comparison algorithms. These algorithms are tested on one simulated and six real images. Meanwhile, all algorithms are executed on the same experimental platform to ensure fairness. Specifically, the experimental system environment is configured with CPU Intel Core i5-12400F CPU@2.5 GHz, 16G memory, Win10 64-bit operating system, and MATLAB simulation platform. Sections III-A–III-G detail the experimental datasets, comparison algorithms and evaluation metrics, parameter analysis, experimental results and analysis, performance of initialization and superpixel generation, computational complexity analysis, and running time of the algorithm.

A. Experimental Datasets

For experimental evaluation, one simulated and six real datasets are selected. Fig. 5 shows the simulated image.

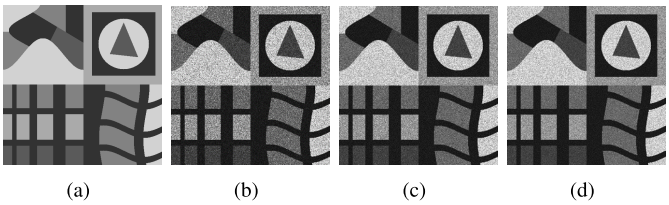


Fig. 5. Simulated SAR images. (a) SI1, (b) 2-look of SI1, (c) 4-look of SI1, and (d) 6-look of SI1.

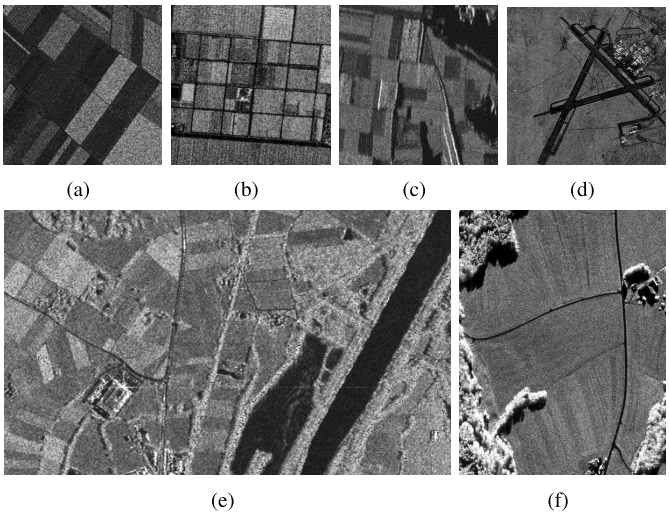


Fig. 6. Real SAR images. (a) Noerdlinger image, (b) Maricopa image, (c) XIAN image, (d) Chinalake image, (e) Rosenheim image, and (f) Traunstein image.

Fig. 5(a) shows the original composite image, referred to as SI1, with dimensions of 512 × 512. This image is significantly complex, containing a mixture of straight lines and curved image regions. It is divided into five classes, corresponding to gray levels of 50, 90, 130, 170, and 210. To simulate real-world SAR imagery, multiplicative speckle noise is artificially introduced into the SI1 image, generating a set of simulated SAR images. By adjusting the noise levels, corresponding 2-look, 4-look, and 6-look simulated SAR images are generated, as shown in Fig. 5(b)–(d). The smaller the number of looks, the more the speckle noise intensifies, leading to increasingly noisy and distorted images.

In addition, six real datasets are evaluated: Noerdlinger, Maricopa, XIAN, Chinalake, Rosenheim, and Traunstein SAR images, as depicted in Fig. 6.

The Noerdlinger dataset captures farmland in the Swabian Jura region of southwestern Germany, with a resolution of 1 m and dimensions of 256 × 256 pixels, divided into four categories. The Maricopa dataset, acquired in the Ku band at the Maricopa Agricultural Center near Arizona, USA, has a resolution of 1 m and dimensions of 350 × 350 pixels. It comprises four categories, including three different types of farmlands and water bodies. The minor variations between regions and significant noise interference may result in uneven segmentation. The XIAN dataset captures farmland in Xi'an, China, using X-band imagery, with a resolution of 1 m and dimensions of 256 × 256 pixels, categorized into four classes.

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The Chinalake dataset, collected using the Ku-band system, covers the China Lake Airport in Ridgecrest, California, USA. It features a resolution of 3 m and dimensions of 446×450 pixels, categorized into three classes: runway, vacant areas, and airport buildings. Two large-scale datasets, Rosenheim and Traunstein, are selected to further validate the algorithm's effectiveness. Both datasets contain a variety of irregular targets, posing significant challenges for segmentation. The Rosenheim dataset, acquired over Rosenheim, Germany, has a size of 831 × 1401 pixels and includes four categories. This dataset is heavily affected by speckle noise. The Traunstein dataset, captured using F-SAR in the X-band over Traunstein in Bavaria, Germany, has a resolution of 1 m and dimensions of 1001 × 779 pixels. It is categorized into three classes: roads, trees, and farmlands.

These datasets encompass various land cover types, such as farmland, trees, roads, buildings, and rivers. They comprehensively evaluate the adaptability and performance of each compared algorithm across diverse ground objects. Moreover, these datasets are applied to practical scenarios to ensure the practical feasibility of the research results.

B. Comparison Algorithms and Evaluation Metrics

Six representative algorithms are adopted for comparison. These algorithms are the fast-weighted CRF (FWCRF) [17], kernel FCM algorithm with pixel intensity and location information (ILKFCM) [40], nonlocal fuzzy clustering algorithm with between-cluster separation measure (NSFCM) [41], superpixel-based fast fuzzy C-means (FCM) clustering (SFFCM) [16], superpixel boundary-based edge description (SpBED) [37], and generalized gamma mixture model superpixel (MISP-GΓD) [22]. The FWCRF, ILKFCM, and NSFCM algorithms are pixel-based SAR image segmentation methods. Conversely, the SFFCM, SpBED, MISP-GΓD, and the proposed RSASGEC are superpixel-based methods.

The segmentation results are presented and analyzed from qualitative and quantitative perspectives. Qualitative analysis involves visual inspection of segmentation result images, while quantitative analysis utilizes five specific performance metrics: segmentation accuracy (SA), precision, recall, $F1$-score, and Kappa.

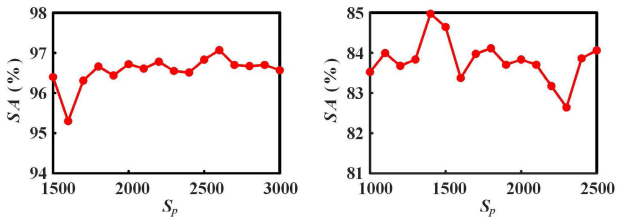
SA is used to assess accuracy by measuring the correspondence between segmentation results and real labels. It calculates the proportion of correctly classified pixels to the total number of pixels

$$\mathrm{SA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{22}$$

where TP is the number of true positives, TN is the number of true negatives, FP is the number of false positives, and FN is the number of false negatives.

The precision represents the proportion of samples correctly classified as positive examples by the algorithm to all samples correctly classified as positive examples by the algorithm. The formula of precision is

$$\mathrm{precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}. \tag{23}$$



(a) (b)

Fig. 7. SA of (a) Noerdlinger and (b) XIAN on different S_p.

The recall represents the proportion of samples correctly classified as positive examples by the algorithm to the actual positive examples. The recall can be computed as follows:

$$\mathrm{recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{24}$$

$F1$-score is the harmonic mean of recall and precision, providing a balanced measure of precision and recall

$$F1\text{-score} = 2 \times \frac{\mathrm{recall} \times \mathrm{precision}}{\mathrm{recall} + \mathrm{precision}}. \tag{25}$$

Meanwhile, the Kappa coefficient is a statistical measure emphasizing consistency in segmentation results compared to random classification. The formula of Kappa is as follows:

$$\mathrm{Kappa} = \frac{\mathrm{SA} - p_e}{1 - p_e} \tag{26}$$

where p_e is the expected classification accuracy, calculated as follows:

$$p_e = \frac{(\mathrm{TP} + \mathrm{FN}) \times (\mathrm{TP} + \mathrm{FP})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})^2} + \frac{(\mathrm{FP} + \mathrm{TN}) \times (\mathrm{FN} + \mathrm{TN})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})^2}. \tag{27}$$

C. Parameter Analysis

1) Influence of the Initial Seed Number S_p: In the RSASGEC algorithm, the initial seed number S_p is a critical parameter determining the final number of generated superpixels, directly influencing the distribution density and shape of the superpixels. Experiments are conducted on the Noerdlinger and XIAN real SAR images to select the optimal S_p. For these two datasets, the ranges of S_p are [1500, 3000] and [1000, 2500], respectively, with an increment of 100. The experimental results are illustrated in Figs. 7 and 8.

Fig. 7 demonstrates the impact of S_p on SA for the Noerdlinger and XIAN datasets, respectively. It can be observed that when S_p is small or large, SA is relatively low. This is attributed to the fact that when S_p is small, the generated superpixels are larger, resulting in insufficient expression of image detail. However, when S_p is large, it may result in oversegmentation, potentially leading to the loss of structural and semantic information.

Fig. 8 demonstrates the impact of S_p on Kappa for the Noerdlinger and XIAN datasets.

It can be seen that the variation trend of Kappa with S_p is consistent with that of SA. Kappa is relatively low when S_p is

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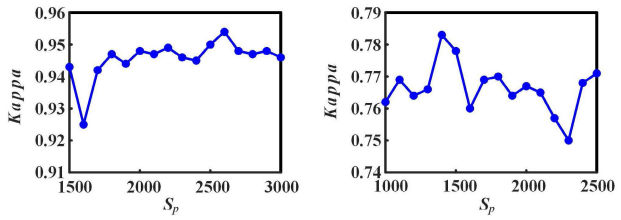


Fig. 8. Kappa of (a) Noerdlinger and (b) XIAN on different S_p.

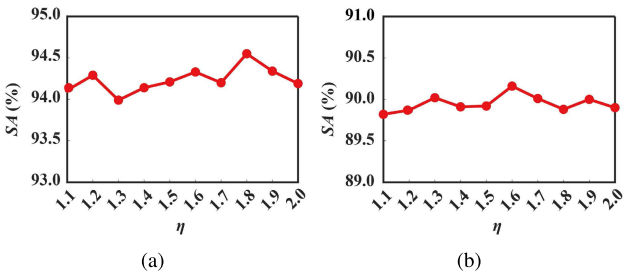


Fig. 9. SA of (a) Chinalake and (b) Traunstein on different η values.

large or small. When S_p is small, each superpixel corresponds to a large region, which may result in blurred boundaries between adjacent areas. Conversely, when S_p is large, it leads to oversegmentation, introducing more complex boundaries between neighboring regions, thus reducing the consistency of segmentation results. In summary, choosing an appropriate S_p is critical for balancing consistency and accurately representing image details. In the following experiments, S_p for the Noerdlinger and XIAN datasets is chosen as 2600 and 1400, respectively.

2) Influence of η on Superpixel Adaptive Energy Function: η is a critical parameter that regulates the distribution of feature weights in an exponential form, influencing the model's ability to differentiate between categories. When η is relatively small, the model's response to feature differences is smoother, grouping superpixels with similar features into the same category. As η increases, the model becomes more sensitive to subtle feature variations, resulting in more precise segmentation. However, if η is set too large, the model may become overly sensitive to noise or outliers, which can degrade segmentation performance. To examine the effect of different η values on the experimental results, we set η within the range of 1.1 to 2, with a step size of 0.1. The Chinalake and Traunstein datasets are used for analysis, and the experimental results for each dataset at different η values are illustrated in Fig. 9.

As shown in Fig. 9(a), for the Chinalake dataset, SA reaches its maximum when η is set to 1.8. Similarly, Fig. 9(b) indicates that for the Traunstein dataset, SA attains a maximum value when η is 1.6. This phenomenon implies that different datasets may have different optimal values of η for achieving the highest SA. Furthermore, the overall trends of SA for the two datasets varying with η are relatively similar. They fluctuate within a specific range, and there is no obvious monotonically increasing or decreasing trend. In addition, the change rate of SA between adjacent η values is relatively small, indicating that SA is less sensitive to the change of η.

TABLE I
QUANTITATIVE RESULTS ON SI1

Algorithm	2-look		4-look		6-look	
	$SA(\%)$	$Kappa$	$SA(\%)$	$Kappa$	$SA(\%)$	$Kappa$
FWCRF [17]	80.97	0.742	92.94	0.907	96.35	0.952
ILKFCM [40]	88.67	0.853	88.89	0.856	89.09	0.859
NSFCM [41]	88.71	0.852	91.01	0.882	92.22	0.898
SFFCM [16]	95.97	0.947	98.28	0.977	98.42	0.979
SpBED [37]	98.56	0.981	98.87	0.985	98.99	0.987
MISP-GΓD [22]	98.64	0.982	99.46	0.993	99.65	**0.995**
RSASGEC	**99.28**	**0.991**	**99.55**	**0.994**	**99.66**	**0.995**

D. Experimental Results and Analysis

Segmentation results of all algorithms are presented and analyzed in this section. Next, the experimental results are analyzed qualitatively and quantitatively.

1) Segmentation Results and Analysis of Simulated Image: In this section, the segmentation results of SI1 simulated images are introduced and analyzed. The compared algorithms' quantitative results on the SI1 images are listed in Table I.

In Table I, the segmentation performance of some comparison algorithms on SI1 is notably reduced, particularly for the FWCRF, ILKFCM, and NSFCM algorithms, where the SA drops below 90%. The performance degradation in FWCRF is mainly attributed to pixel smoothing, resulting in a significant gray difference between the current and previous pixels. Moreover, the FWCRF algorithm exhibits less robustness, with SA for the 2-look image significantly lower than that of the 4-look image as noise increases. Notably, the 2-look image has higher noise than the 4-look image. Among the superpixel-based methods, SFFCM achieves relatively strong segmentation accuracy only on low-noise images. The SA for the 4-look and 6-look images exceeds 98%, while the 2-look image reaches only 95.97%. Other superpixel-based methods, including SpBED, MISP-GΓD, and the proposed RSASGEC, demonstrate promising segmentation performance. Specifically, the proposed RSASGEC effectively segments multiple targets in complicated regions by adaptively treating homogeneous and complicated regions in different ways.

The results on the 2-look SI1 image are shown in Fig. 10. In Fig. 10, the SI1 image exhibits straight lines and curves, with more corner points and closer pixel gray levels, making segmentation more challenging. In Fig. 10(c)–(e), numerous speckles appear in homogeneous areas, particularly in Fig. 10(c), where intraclass consistency is weaker compared to Fig. 10(f)–(h). This difference arises because the algorithms in Fig. 10(f)–(h) are superpixel-based, taking spatial context information into account, which effectively retains the image's structural information. However, as shown in Fig. 10(h), MISP-GΓD exhibits obvious segmentation errors in the green and red areas. In Fig. 10(g), SpBED has superpixel segmentation errors, resulting in clustering errors and small misclassification areas at boundary edges. The proposed RSASGEC effectively segments high-complexity superpixel

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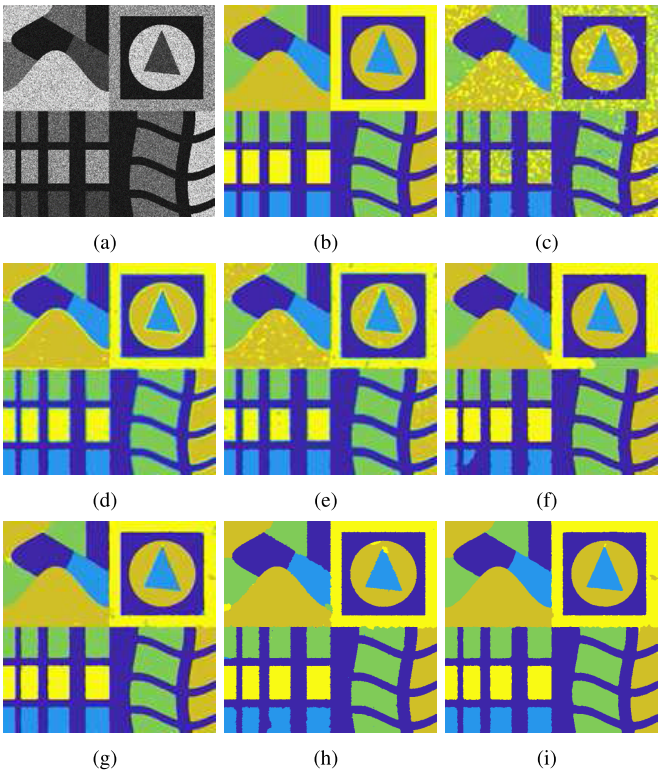


Fig. 10. Experimental results of each algorithm on the 2-look SI1 simulated image. (a) 2-look SI1, (b) Ground truth, (c) FWCRF, (d) ILKFCM, (e) NSFCM, (f) SFFCM, (g) SpBED, (h) MISP-GΓD, and (i) RSASGEC.

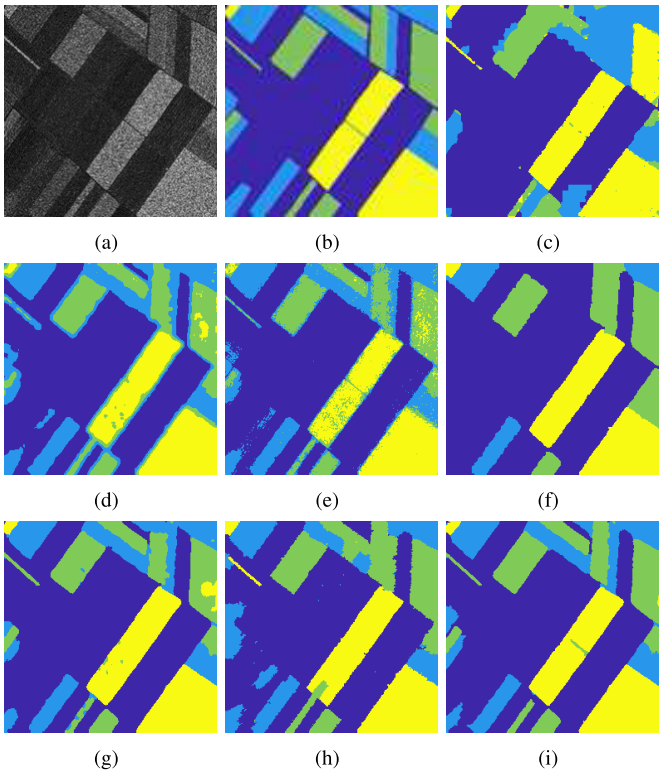


Fig. 11. Segmentation results of the Noerdlinger image. (a) Noerdlinger image, (b) Ground truth, (c) FWCRF, (d) ILKFCM, (e) NSFCM, (f) SFFCM, (g) SpBED, (h) MISP-GΓD, and (i) RSASGEC.

TABLE II
QUANTITATIVE RESULTS OF THE NOERDLINGER IMAGE

Algorithm	$SA(\%)$	$Kappa$	$precision(\%)$	$recall$	$F1\text{-}score$
FWCRF [17]	77.47	0.644	46.40	0.409	0.435
ILKFCM [40]	87.79	0.814	87.34	0.623	0.727
NSFCM [41]	90.56	0.852	87.09	0.721	0.789
SFFCM [16]	87.19	0.787	42.66	**0.940**	0.587
SpBED [37]	94.66	0.916	87.95	0.872	0.876
MISP-GΓD [22]	92.85	0.888	79.63	0.817	0.806
RSASGEC	**97.10**	**0.955**	**93.28**	0.921	**0.927**

regions, making the superpixel segmentation region more compact, as shown in Fig. 10(i). This is primarily attributed to our design of a complicated region-sensitive adaptive superpixel generation algorithm, alleviating the issue of superpixel segmentation errors.

2) Segmentation Results and Analysis of Real Images: The tests of all five metrics on Noerdlinger, Maricopa, XIAN, Chinalake, Rosenheim, and Traunstein real SAR datasets are conducted. Quantitative analyses are conducted to evaluate the performance of the RSASGEC and the compared algorithms. The quantitative results of the Noerdlinger image are listed in Table II.

In Table II, the RSASGEC algorithm achieves superior performance on the Noerdlinger dataset, achieving an SA of 97.10% and a Kappa of 0.955. This outperforms the suboptimal SpBED algorithm, which attains 94.66% and 0.916 on these metrics. Additionally, RSASGEC exhibits a substantial precision improvement, rising from 87.95% to 93.28%, an increase of 5.33%. This indicates enhanced accuracy in predicting positive instances. The RSASGEC algorithm's success is attributed to its initialization based on the complexity of regions, effectively distinguishing between complex and homogeneous regions. Furthermore, the $F1$-score reaches 0.927, surpassing the next best algorithm's 0.876, proving the superior balance between precision and recall of the RSASGEC algorithm.

The results of the Noerdlinger image are shown in Fig. 11.

In Fig. 11(c)–(e), the FWCRF, ILKFCM, and NSFCM algorithms demonstrate poor consistency in complex regions. Specifically, in Fig. 11(e), the yellow farmland contains numerous green pixels, highlighting the challenges in accurately segmenting complex regions. In contrast, the superpixel-based SFFCM, SpBED, and MISP-GΓD algorithms in Fig. 11(f)–(h) show improved segmentation capabilities in handling these complex regions, indicating that superpixel-based algorithms can more effectively preserve complex structural information of targets. However, due to the lack of utilization of edge information, the boundary adherence of the SFFCM and MISP-GΓD algorithms is less than ideal. The SpBED algorithm, on the other hand, achieves satisfactory boundary adherence by leveraging strong edge information. However, SpBED employs regularized initialization for superpixel generation, lacking the ability to distinguish between complex and homogeneous regions, resulting in less accurate pixel segmentation in some complex areas. In comparison, the proposed RSASGEC algorithm produces smoother and more compact segmentation results, as shown in Fig. 11(i). This is primarily attributed to considering region complexity during superpixel initialization and employing an adaptive

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TABLE III
QUANTITATIVE RESULTS OF THE MARICOPA IMAGE

Algorithm	$SA(\%)$	$Kappa$	$precision(\%)$	$recall$	$F1\text{-}score$
FWCRF [17]	69.46	0.569	49.08	0.677	0.569
ILKFCM [40]	66.67	0.533	43.82	0.534	0.481
NSFCM [41]	65.00	0.519	70.31	0.657	0.679
SFFCM [16]	77.36	0.674	66.64	0.762	0.711
SpBED [37]	84.29	0.774	86.26	0.788	0.824
MISP-GΓD [22]	78.49	0.690	83.31	0.667	0.741
RSASGEC	**91.92**	**0.883**	**91.54**	**0.884**	**0.899**

initialization approach which handles regions differently depending on their complexity. Additionally, edge correction enables the effective preservation of edge information in structurally diverse regions.

The results of the Maricopa image are listed in Table III.

As shown in Table III, the proposed RSASGEC algorithm achieves the best performance on all five metrics for the Maricopa dataset. The SA on the Maricopa image reaches 91.92%, and Kappa reaches 0.883. Compared with the less optimal algorithm SpBED, our proposed RSASGEC algorithm improves by 7.63% and 0.109 in these two metrics respectively. This improvement can be attributed to the RSASGEC algorithm considering the impact of speckle noise throughout the entire process, effectively mitigating the adverse effects of speckle noise on SAR image segmentation performance. Simultaneously, the RSASGEC algorithm also shows a significant improvement in precision, recall, and $F1$-score compared to the next best algorithms, with increases of 5.28%, 0.096, and 0.075, respectively. This superiority is attributed to the adaptive energy function proposed in this article, which introduces feature weights. By adaptively adjusting the weights of different features based on local image features, the energy function becomes more flexible, and the generated superpixels are more compact. These advantages indicate that the RSASGEC algorithm has more robust segmentation capabilities, especially when dealing with complex SAR images.

The results of the Maricopa image are shown in Fig. 12. In Fig. 12(c)–(e), serious noise is observed in the homogeneous region. These images exhibit numerous cyan pixels in green fields, indicating that the segmentation ability for complex images still requires improvement. FWCRF shows particularly severe issues. It performs smoothing operations, rebalancing the gray levels between adjacent pixels and leading to segmentation errors. The superpixel-based algorithms in Fig. 12(f)–(h) demonstrate relatively superior segmentation performance for complex regions. The SFFCM and MISP-GΓD algorithms neglect edge information, inaccurately dividing many pixel blocks into adjacent areas. SpBED utilizes multiscale and multidimensional Gabor edge detection windows to identify image boundaries, effectively preventing errors in edge pixel segmentation. The RSASGEC can more effectively segment boundaries and complex regions, as shown in Fig. 12(i). This is attributed to the adaptive, nonuniform initialization of superpixel seeds, based on the region complexity.

The results of the XIAN image are listed in Table IV.

Table IV demonstrates that the RSASGEC algorithm outperforms all other algorithms on the XIAN image. The SA

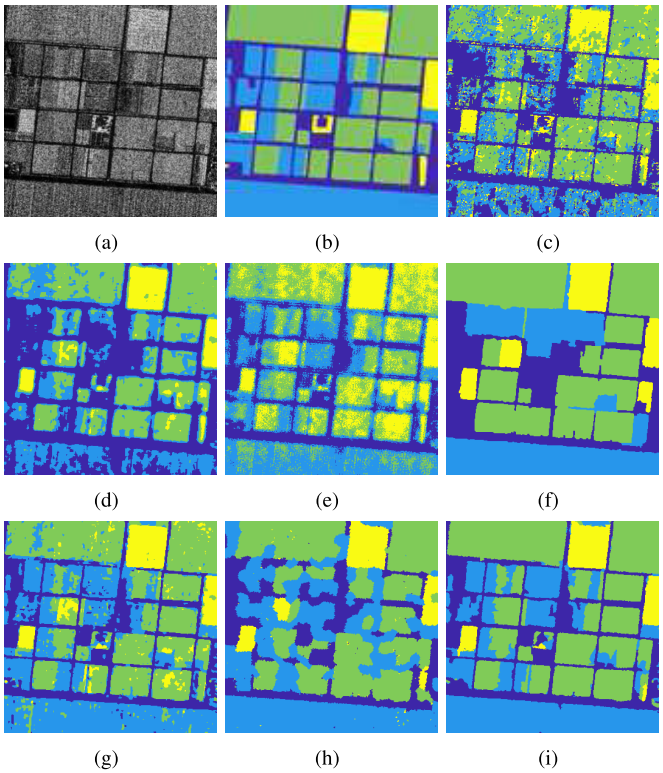


Fig. 12. Segmentation results of the Maricopa image. (a) Maricopa image, (b) Ground truth, (c) FWCRF, (d) ILKFCM, (e) NSFCM, (f) SFFCM, (g) SpBED, (h) MISP-GΓD, and (i) RSASGEC.

TABLE IV
QUANTITATIVE RESULTS OF THE XIAN IMAGE

Algorithm	$SA(\%)$	$Kappa$	$precision(\%)$	$recall$	$F1\text{-}score$
FWCRF [17]	66.39	0.524	52.40	0.520	0.522
ILKFCM [40]	60.64	0.461	27.78	0.368	0.317
NSFCM [41]	67.04	0.558	74.21	0.782	0.762
SFFCM [16]	44.09	0.305	36.52	0.656	0.469
SpBED [37]	82.40	0.750	74.54	0.840	0.790
MISP-GΓD [22]	80.48	0.724	75.20	0.729	0.740
RSASGEC	**85.02**	**0.786**	**77.01**	**0.873**	**0.818**

and Kappa of the RSASGEC are 2.62% and 0.036 higher than those of the next best comparison algorithm. Moreover, RSASGEC achieves the best precision, recall, and $F1$-score, with improvements of 2.47%, 0.033, and 0.028, respectively, compared to the next best algorithm. This superior performance is mainly attributed to RSASGEC's consideration of the irregular shape and sensitivity to noise in the XIAN image. The noise impact is mitigated through comprehensive preprocessing, superpixel generation, and postprocessing stages. By employing a model based on Gaussian edge detection and the Canny algorithm for edge correction, RSASGEC demonstrates enhanced speckle noise suppression. Additionally, the design of complicated region-sensitive dynamic initialization effectively addresses segmentation challenges in these complex image regions.

The segmentation results of the XIAN image are shown in Fig. 13.

In Fig. 13(c)–(e), numerous speckles are observed in homogeneous regions, which is the characteristic of pixel-based

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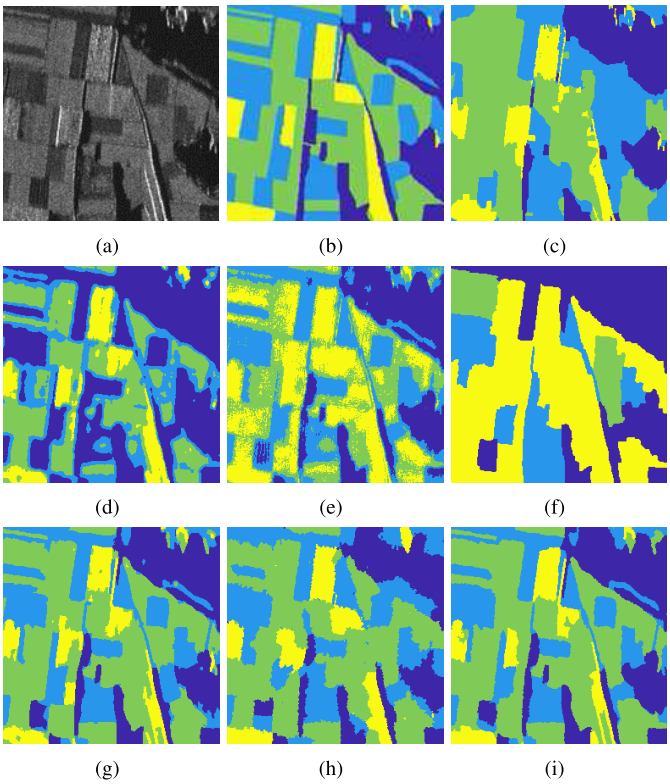


Fig. 13. Segmentation results of the XIAN image. (a) XIAN image, (b) Ground truth, (c) FWCRF, (d) ILKFCM, (e) NSFCM, (f) SFFCM, (g) SpBED, (h) MISP-GΓD, and (i) RSASGEC.

algorithms which overlook spatial context. Consequently, these algorithms struggle to differentiate noise from the gray mutation in the homogeneous regions. Conversely, Fig. 13(f)–(h) demonstrates improved segmentation capabilities in complex regions. For instance, SFFCM, shown in Fig. 13(f), fails to segment the entire image into smaller regions using boundary information, leading to inaccuracies in many areas, despite successfully avoiding speckles in the homogeneous regions. In contrast, SpBED, shown in Fig. 13(g), achieves high segmentation accuracy for the water body region but encounters challenges in correcting the farmland region's errors in homogeneous regions, resulting in reduced overall accuracy. However, as shown in Fig. 13(h), the overall segmentation performance is satisfactory, offering more accurate segmentation of farmland and water, thereby achieving higher accuracy. The RSASGEC demonstrates higher accuracy in segmenting cyan and dark blue regions and preserves edge information better in light blue regions, as seen in Fig. 13(i).

The results of the Chinalake image are listed in Table V. As shown in Table V, the RSASGEC algorithm demonstrates good segmentation performance on the Chinalake dataset, with SA, Kappa, precision, and $F1$-score reaching 94.66%, 0.829, 95.88%, and 0.968, respectively. In comparison, SpBED exhibits suboptimal results on these metrics, with RSASGEC improving by 1.71%, 0.044, 2.42% and 0.011, respectively. However, SpBED performs best in terms of recall, improving from 0.977 to 0.980 over RSASGEC. Compared with the pixel-based FWCRF, ILKFCM, and NSFCM methods,

TABLE V
QUANTITATIVE RESULTS OF THE CHINALAKE IMAGE

Algorithm	$SA(\%)$	$Kappa$	$precision(\%)$	$recall$	$F1\text{-}score$
FWCRF [17]	54.27	0.257	48.42	0.919	0.634
ILKFCM [40]	59.23	0.303	53.61	0.940	0.683
NSFCM [41]	59.88	0.324	53.16	0.958	0.666
SFFCM [16]	60.06	0.325	53.39	0.964	0.687
SpBED [37]	92.95	0.785	93.46	**0.980**	0.957
MISP-GΓD [22]	88.69	0.678	89.58	0.971	0.932
RSASGEC	**94.66**	**0.829**	**95.88**	0.977	**0.968**

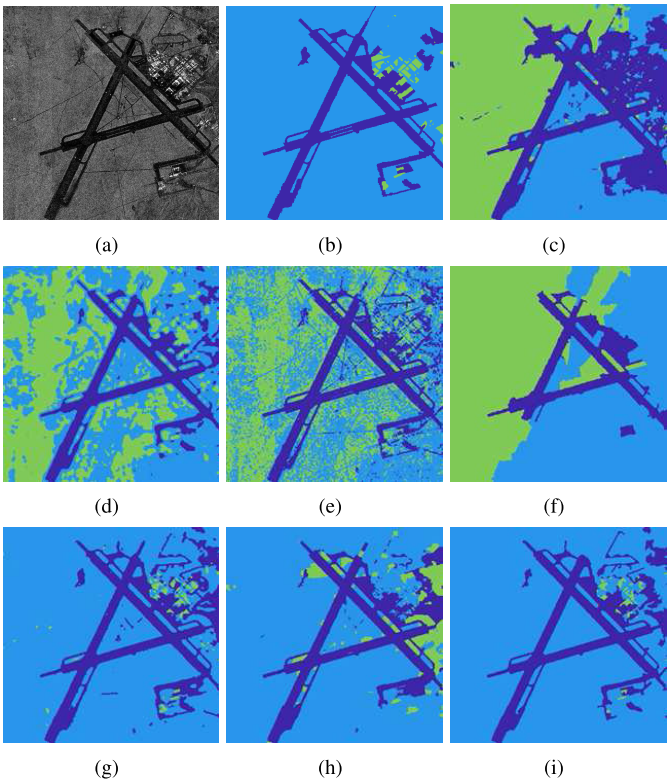


Fig. 14. Segmentation results of the Chinalake image. (a) Chinalake image, (b) Ground truth, (c) FWCRF, (d) ILKFCM, (e) NSFCM, (f) SFFCM, (g) SpBED, (h) MISP-GΓD, and (i) RSASGEC.

the superpixel-based SFFCM, SpBED, MISP-GΓD, and the proposed algorithm achieve better segmentation performance. Specifically, the pixel-based methods generally achieve an SA of around 60%. Among the superpixel-based algorithms, the SA of the SFFCM algorithm is only slightly above 60%. The SA of the SpBED and RSASGEC algorithms is above 90% due to their effective use of spatial information and good noise and boundary region processing, which is superior to the other algorithms.

The segmentation results of Chinalake are shown in Fig. 14.

In Fig. 14, the outputs of the compared algorithms for SAR image segmentation are illustrated. Specifically, the FWCRF, ILKFCM, and NSFCM are pixel-based methods, as shown in Fig. 14(c)–(e). Conversely, the SFFCM, SpBED, MISP-GΓD, and RSASGEC are superpixel-based methods, depicted in Fig. 14(f)–(i). It is evident that the segmentation performance of the pixel-based methods is inferior to that of the superpixel methods. In Fig. 14, green, dark blue, and light blue areas represent airport buildings, runways, and vacant areas. The

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TABLE VI
QUANTITATIVE RESULTS OF THE ROSENHEIM IMAGE

Algorithm	$SA(\%)$	$Kappa$	$precision(\%)$	$recall$	$F1\text{-}score$
FWCRF [17]	66.94	0.509	87.75	0.632	0.735
ILKFCM [40]	62.59	0.493	51.55	0.841	0.639
NSFCM [41]	66.31	0.537	62.89	0.813	0.709
SFFCM [16]	66.02	0.445	66.03	0.761	0.707
SpBED [37]	70.42	0.587	73.93	0.822	0.779
MISP-GΓD [22]	69.90	0.581	71.75	0.824	0.767
RSASGEC	**82.41**	**0.751**	**86.47**	**0.865**	**0.865**

results of the pixel-based methods reveal numerous speckles. There is a lot of noise in the vacant area of FWCRF, and it is misclassified as an airport building, while the airport building is misidentified as a runway. This is mainly due to the smoothing operation of the FWCRF algorithm, which causes gray-level convergence among superpixel center points and adjacent regions. In contrast, superpixel-based algorithms produce segmentation results with greater intraclass consistency, demonstrating their effectiveness in maintaining pixel consistency. Although the SpBED and MISP-GΓD algorithms can segment the image of Chinalake accurately, there are still a few spots in complex areas. RSASGEC achieves more promising segmentation performance, mainly due to the complex region-sensitive dynamic initialization optimization process.

The results of the Rosenheim are listed in Table VI.

As shown in Table VI, the RSASGEC performs best on the Rosenheim dataset, achieving SA and Kappa of 82.41% and 0.751, respectively. RSASGEC surpasses the next best algorithm by 11.99% and 0.164 on these metrics. Meanwhile, the RSASGEC algorithm achieves the best precision, recall, and $F1$-score, with improvements of 12.54%, 0.043, and 0.086, respectively, compared to the next best algorithm. Notably, Noerdlinger exhibits the best segmentation performance, while Rosenheim demonstrates the worst. This discrepancy is attributed to Noerdlinger's uniform distribution, while Rosenheim exhibits a nonuniform distribution with distinct homogeneous and complex regions and is heavily affected by speckle noise. RSASGEC can alleviate these challenges by employing the complicated region-sensitive dynamic initialization method. By dynamically partitioning superpixels based on region complexity and variations, RSASGEC allocates more superpixel seeds in complex regions, and fewer in homogeneous regions, prior to local clustering. This design makes the generated superpixels more adaptable to the objects in the image, thus improving the performance.

The segmentation results of the Rosenheim image are shown in Fig. 15. Compared to the other five datasets, the results on the Rosenheim dataset are generally weaker. In Fig. 15(c)–(e), numerous speckles are observed in the homogeneous regions, which is the characteristic of pixel-based algorithms that do not consider the spatial context. Consequently, these algorithms struggle to differentiate noise in gray mutation from the homogeneous regions effectively. In Fig. 15(f)–(h), the superpixel-based algorithms exhibit improved segmentation capabilities for complex regions. For

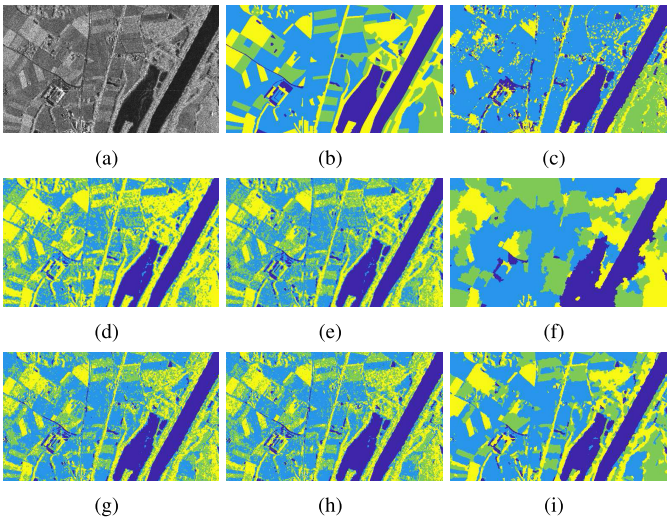


Fig. 15. Segmentation results of the Rosenheim image. (a) Rosenheim image, (b) Ground truth, (c) FWCRF, (d) ILKFCM, (e) NSFCM, (f) SFFCM, (g) SpBED, (h) MISP-GΓD, and (i) RSASGEC.

TABLE VII
QUANTITATIVE RESULTS OF THE TRAUNSTEIN IMAGE

Algorithm	$SA(\%)$	$Kappa$	$precision(\%)$	$recall$	$F1\text{-}score$
FWCRF [17]	85.77	0.652	96.67	**0.968**	**0.967**
ILKFCM [40]	88.27	**0.757**	97.82	0.862	0.917
NSFCM [41]	88.21	0.670	99.47	0.885	0.937
SFFCM [16]	84.78	0.550	99.44	0.853	0.918
SpBED [37]	89.63	0.720	98.99	0.905	0.945
MISP-GΓD [22]	89.50	0.703	99.37	0.896	0.943
RSASGEC	**90.16**	0.734	**99.48**	0.909	0.950

instance, SFFCM, as depicted in Fig. 15(f), fails to segment the entire image into smaller regions using boundary information, resulting in incorrect segmentation of many areas, even though it successfully avoids speckles in the homogeneous region. The SpBED, illustrated in Fig. 15(g), achieves high segmentation accuracy for the water body region but encounters challenges in correcting the farmland region's error in the homogeneous region, resulting in reduced overall accuracy. However, as shown in Fig. 15(h), the overall segmentation performance is satisfactory, offering more accurate segmentation of farmland and water, thereby achieving higher accuracy. The RSASGEC demonstrates higher accuracy in segmenting blue regions and the segmentation result is relatively pure, as seen in Fig. 15(i).

The results of the Traunstein are listed in Table VII.

As shown in Table VII, the SA of all algorithms exceeds 80% when dealing with the relatively low-noise Traunstein dataset. The performance gap between the compared algorithms is relatively small, indicating that these algorithms can all effectively process low-noise datasets. The RSASGEC algorithm has demonstrated excellent performance on many metrics. Specifically, its SA and precision reach 90.16% and 99.48%, respectively. Compared with the next best algorithm, RSASGEC improves by 0.53% and 0.01%, respectively. In particular, the Kappa of ILKFCM is the highest, indicating that ILKFCM has a more balanced segmentation ability for different targets. The FWCRF algorithm performs best

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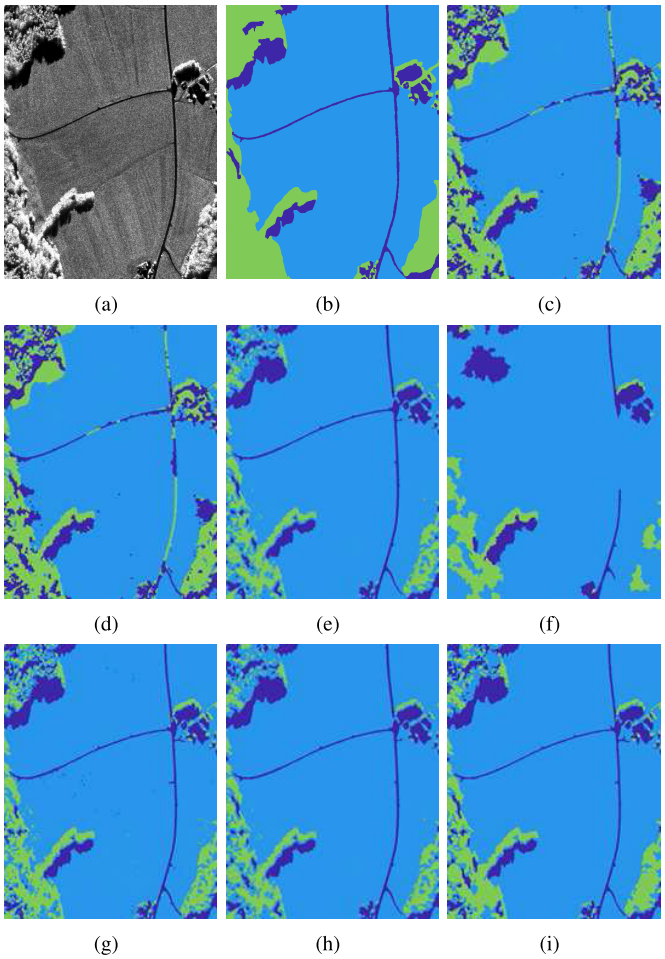


Fig. 16. Segmentation results of the Traunstein image. (a) Traunstein image, (b) Ground truth, (c) FWCRF, (d) ILKFCM, (e) NSFCM, (f) SFFCM, (g) SpBED, (h) MISP-GΓD, and (i) RSASGEC.

on recall and can identify more actual positive samples. However, it is slightly worse in SA, Kappa, and precision. This may be attributed to the limitations of the FWCRF algorithm in processing spatial relations in images. Although the conditional random field (CRF) model can consider spatial relations between pixels, it is still affected by factors such as texture changes when confronted with complex and diverse SAR images, leading to mis-segmentation. In conclusion, the RSASGEC algorithm shows excellent performance and high consistency on the Traunstein dataset. Its adaptive superpixel generation and edge correction strategies provide a strong segmentation ability.

The segmentation results of the Traunstein image are shown in Fig. 16.

The green, dark blue, and light blue areas represent trees, roads, and fields. It is evident that the segmentation result of the SFFCM algorithm is least satisfactory, as shown in Fig. 16(f). Specifically, the SFFCM algorithm fails to identify roads, and incorrectly categorizes numerous trees as farmland. This limitation can be attributed to the algorithm's excessive reliance on the color features of superpixels during the clustering process. By overlooking the subtle distinctions between small targets and their surrounding environments, the SFFCM algorithm struggles to segment these features effectively.

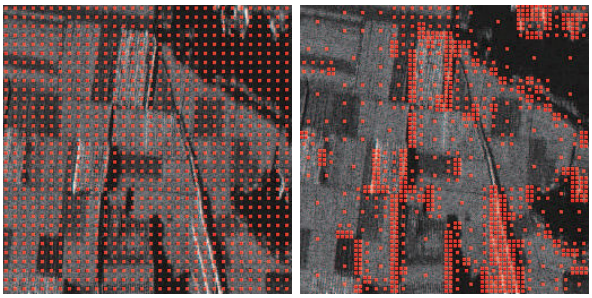


Fig. 17. Initialization results. (a) Results of SLIC-GC and (b) results of the RSASGEC.

In Fig. 16(c), the FWCRF result contains numerous misclassified pixels, primarily due to smoothing operations, which may obscure essential details, making it difficult to distinguish regions with low gray-scale differences. For the ILKFCM and NSFCM shown in Fig. 16(d) and (e), the pixel classification in boundary regions is ambiguous, with some misclassifications occurring. The boundary adherence is poor due to the lack of consideration for edge information. Conversely, the SpBED and RSASGEC methods demonstrate superior segmentation performance by incorporating edge information, as illustrated in Fig. 16(g) and (i).

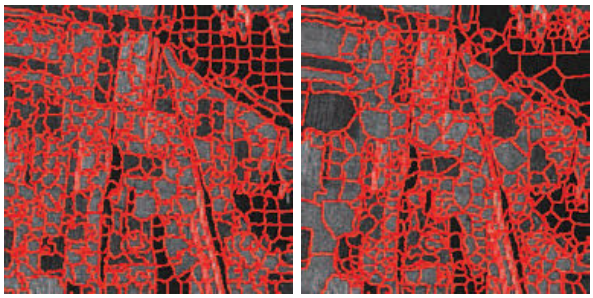
E. Performance of Initialization and Superpixel Generation

The initialization and superpixel generation methods play a crucial role in superpixel segmentation. To validate the effectiveness of the initialization and superpixel generation algorithm, SLIC-GC is compared with the algorithm in this article. SLIC-GC adopts a grid-centered initialization strategy, where initial seeds are selected based on regular grid partitioning of the image. In contrast, our proposed initialization method dynamically partitions the image based on the complexity of image regions. The XIAN dataset contains both homogeneous and complex regions, making it suitable for this comparative experiment. To ensure fairness, the number of superpixels is set to 600. The similarity function is consistent with the proposed algorithm. The superpixel initialization results for SLIC-GC and RSASGEC are shown in Fig. 17.

Fig. 17 shows superpixel initialization results for SLIC-GC and the RSASGEC algorithm. In Fig. 17(a), it can be seen that the initial superpixel seeds of SLIC-GC exhibit a relatively uniform distribution. In contrast, Fig. 17(b) reveals the uneven distribution of the initial superpixel seeds of RSASGEC, which are more dense in complex regions and more sparse in homogeneous regions.

To further demonstrate the effectiveness of our proposed method in generating superpixels, the superpixel results before the postprocessing step are shown in Fig. 18. Fig. 18 shows the different morphologies of the superpixels generated by the SLIC-GC and RSASGEC algorithms on the XIAN dataset. In Fig. 18(a), it can be observed that the superpixels generated by SLIC-GC exhibit a relatively uniform distribution. The superpixels maintain a regular grid-like pattern, especially in the homogeneous region in the upper right corner. In contrast,

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(a) (b)

Fig. 18. Superpixel results. (a) Results of SLIC-GC and (b) results of the RSASGEC.

Fig. 18(b) reveals the uneven distribution of superpixels for RSASGEC, which are dense in complex regions and more sparse in homogeneous regions. This can be attributed to the dynamic initialization based on image complexity and iterative superpixel segmentation based on adaptive energy function, allowing the generated superpixels to better adapt to the image structure.

F. Computational Complexity Analysis

Assume that the input image size is $M \times N$, and the number of superpixels is L. The computational complexity of the RSASGEC algorithm consists of three main components: irregular initialization of superpixels, iterative superpixel segmentation, and label correction. For superpixel initialization, this article initializes superpixels based on region complexity. First, the image is divided into four superpixels, and their complexity is calculated and sorted. Subsequently, the region with the highest complexity is further subdivided, and this process continues until the specified number of superpixels is reached. The time complexity of this process is $O(L/3 \times M/2 \times N/2)$, approximately $O(L \times M \times N)$. For superpixel generation, an iterative search is performed for each superpixel center within its neighborhood range $2R \times 2R$. Since superpixel generation involves n iterations, the computational complexity is $O(L \times 2R \times 2R \times N) \ll O(L \times M \times N)$. For the label correction, the time complexity is $O(M \times N)$. In summary, the total computational complexity of the proposed RSASGEC algorithm is $O(L \times M \times N)$.

G. Running Time of the Algorithm

To analyze the algorithm's running time, the RSASGEC algorithm and six other comparison algorithms are tested on four real datasets. The running time of the algorithms are shown in Table VIII. From Table VIII, it can be observed that SFFCM has the shortest running time. This is mainly because SFFCM uses a multiscale morphological gradient reconstruction (MMGR) operation and watershed transform to efficiently generate superpixels, which can effectively extract local structure information of images. Additionally, FCM superpixel clustering is carried out using histogram parameters, which avoids the detailed classification of each pixel and significantly reduces the running time. On the contrary, ILKFCM exhibits the longest running time. ILKFCM employs a kernel clustering algorithm with a fuzzy factor, which needs to calculate the similarity between samples in a high-dimensional space, resulting in high computational complexity. Moreover, each iteration needs to calculate the fuzzy factor, increasing the extra computational overhead.

TABLE VIII
TIME COST OF REAL SAR IMAGES

Algorithm	Noerdlinger	Maricopa	XIAN	Rosenheim
	Time(s)	Time(s)	Time(s)	Time(s)
FWCRF [17]	3.15	5.40	3.27	68.42
ILKFCM [40]	52.98	138.22	73.93	1266.24
NSFCM [41]	14.03	26.27	14.43	275.13
SFFCM [16]	**0.39**	**0.54**	**0.44**	**7.20**
SpBED [37]	5.09	9.07	4.79	76.00
MISP-GГD [22]	53.37	349.84	54.68	1072.16
RSASGEC	8.64	13.97	6.29	104.17

The running time of the RSASGEC algorithm is almost equal to that of other algorithms. This indicates that the RSASGEC algorithm can balance both computational efficiency and performance well. In summary, for different segmentation algorithms, the reason for the short running time is often related to the efficient extraction of local structures and the optimization of the clustering process. In this regard, RSASGEC has shown satisfactory overall performance.

IV. CONCLUSION

This article has proposed a complicated RSASGEC algorithm to tackle the challenge of segmenting multiple targets in complex images. Initially, a complicated region-sensitive dynamic initialization method is introduced to better reflect the structure of the image, and improve the discrimination of multiple targets within complex regions. Subsequently, an adaptive energy function was developed, utilizing adaptive feature weights to dynamically balance the influence of gray and spatial features on similarity. A Lagrange multiplier approach is employed within the adjustable search window to optimize the adaptive energy function, resulting in more compact superpixels. To further refine segmentation, the algorithm utilizes Gaussian edge detection and the Canny algorithm to detect edges, providing crucial edge information for subsequent superpixel fusion and label correction. Majority voting and region growth methods with edge constraints are incorporated, to effectively mitigate edge noise and outliers. The RSASGEC algorithm was evaluated on one simulated image and six real SAR images, demonstrating consistently impressive performance. On the simulated images, SA exceeded 99% and Kappa exceeded 0.99. When compared with SpBED on the Maricopa image, the RSASGEC algorithm exhibits improvements of 7.63% and 0.109 in these two metrics. Despite facing challenges in segmenting the nonhomogeneous and noisy Rosenheim image, RSASGEC outperforms the suboptimal SpBED, achieving an 11.99% improvement in SA and a 0.164 improvement in Kappa. Quantitative metrics and visual inspections confirm the generally superior performance of superpixel-based algorithms over pixel-based alternatives, especially when considering edge information. RSASGEC's approach of adaptively initializing superpixel seeds based on region complexity effectively preserves edge details, significantly enhancing the segmentation performance of multiple objects in complex image regions.

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